

Ethiopian Online Lottery Platform

Product & infrastructure plan — multi-tenant lottery SaaS (web app + Telegram bot)

Model: B2B SaaS · Market: Ethiopia · Channels: Web + Telegram per client · Date: July 10, 2026 · Prices verified July 2026

1. At a glance

QUESTION	ANSWER
What we're building	Lottery-as-a-Service: a multi-tenant platform where clients (lottery operators) run their own branded lottery — web app + Telegram bot — on our software
Business model	Subscription (monthly fee) + revenue share or per-ticket fee, billed to clients; player money never touches us
Recommended stack	Laravel + MySQL/PostgreSQL + Redis — one deployment, tenant-scoped from day one
Hosting (pay in birr)	Yegara VPS 80G — 4,450 ETB/mo (4 vCPU, 8 GB RAM, 80 GB NVMe)
Hosting (if we can pay in EUR/USD)	Hetzner Cloud CX32 — ~€7/mo (~1,200 ETB) for the same specs, roughly ¼ the price
Infrastructure running cost	~5,000–7,500 ETB/month to start on the birr-only route; ~2,500–4,000 ETB/month via Hetzner

2. Channels (per tenant)

Every client gets their own branded instance of each channel:

CHANNEL	ROLE
Web app (responsive, PWA)	Full experience: registration, wallet, play, results, account — on the client's subdomain or custom domain
Telegram bot	Full play loop in chat: register, deposit, buy, get notified. In Ethiopia a large share of lottery/betting activity happens on Telegram — each client connects their own bot
Telegram Mini App	The client's web app opened inside Telegram via their bot — same codebase, near-zero extra cost, feels native to Telegram users
Public Telegram channel	Client marketing: auto-posted draw results, winner announcements, jackpot updates
Later (V2+)	Agent/retailer portal for cash sales, USSD (requires telecom agreement), native mobile apps

3. SaaS model: one platform, many lottery operators

We are not building one lottery — we’re building the **system that runs lotteries** and selling it to operators. Three layers:

LAYER	WHO	WHAT THEY DO / GET
Platform (us)	We operate the software	Host, maintain, onboard and bill tenants; never touch player money
Tenants (clients)	Lottery operators	Their own branded lottery: domain, Telegram bot, games, players, prize payouts
Players	Each tenant’s customers	Play on that tenant’s web app/bot; belong to exactly one tenant

3.1 Multi-tenancy architecture

- **One codebase, one deployment, shared database with a tenant_id on every row** (Laravel: stancl/tenancy or spatie/laravel-multitenancy). Simplest to operate on one VPS; scales to dozens of tenants
- **Tenant resolution by domain:** each client gets `client.ourplatform.com` (wildcard DNS + wildcard SSL); custom domains (`clientlottery.com`) supported via CNAME with automated Let’s Encrypt issuance
- **Every query tenant-scoped** through global scopes, enforced by automated tests — a cross-tenant data leak is the one unforgivable bug in a SaaS
- Upgrade path: **database-per-tenant** for premium clients who demand hard isolation (same codebase supports both modes)
- **Per-tenant Telegram bot:** the client creates a bot with BotFather and pastes the token; the platform registers one webhook per bot and routes updates to the right tenant
- **Per-tenant payment credentials:** each client connects their **own** Chapa / SantimPay / ArifPay / Telebirr merchant account. Player deposits settle directly with the client — the platform never holds player funds (keeps us a pure technology provider)
- Per-tenant branding (logo, colors, name, T&C), SMS sender ID and quota, language defaults
- Everything in the feature list (ledger, draw engine, fraud tools, reports) runs tenant-scoped automatically

3.2 Platform admin console (our side — sits above all tenant admins)

- **Tenant lifecycle:** onboarding wizard (brand → bot token → payment keys → first game), trial period, activate, suspend, offboard with full data export
- **Subscription plans & feature flags** per plan: allowed game types, player caps, SMS quotas, custom domain, Mini App, API access
- **Billing:** monthly fee + revenue share (% of ticket sales) or per-ticket fee; usage metered automatically from the ledger; invoices in birr; overdue → warning → auto-suspend
- Cross-tenant health dashboard: sales, error rates, queue depth, draw schedule per tenant
- Support **impersonation** (“view as tenant admin”) — fully audit-logged
- Platform-wide announcements to tenant admins

3.3 Consequences for the rest of this plan

- The admin console in the feature list is the **tenant admin console**; the platform console sits above it
- **Build tenant-aware from day one**, even if we launch with a single anchor tenant (possibly our own lottery operation). Retrofitting `tenant_id` into a live single-tenant money system later is one of the most painful migrations there is
- Each tenant carries their own operating license and prize liabilities — this belongs in the client contract; the platform contracts as a technology provider

4. Feature list (comprehensive — everything below is per tenant)

4.1 Accounts & identity

- Registration with **phone number + SMS OTP** (phone-first; email optional)
- Login: phone + password/PIN; OTP fallback; “login with Telegram” on web
- **One account across web and bot** — link Telegram to an existing account via OTP or deep-link code
- Profile: name, language, notification preferences
- **KYC tiers with limits:**
 - Tier 0 (phone verified): small deposits/plays, no withdrawals
 - Tier 1 (ID verified — Fayda national ID, kebele ID, or passport photo upload + admin review): full limits, withdrawals enabled
- Age verification (18+) at KYC; date-of-birth check at registration
- Transaction PIN and/or 2FA required for withdrawals
- Device & session management (view/revoke active sessions)
- **Responsible gaming:** self-set deposit limits, cool-off periods, self-exclusion, visible spend history
- Account closure with data-retention rules

4.2 Wallet & payments

- **Double-entry ledger** — every birr movement is an immutable journal entry; balances are derived, never edited directly. This is the audit backbone for disputes, reconciliation, and SaaS usage metering
- Single wallet shared across web + bot; separate **cash balance** and **bonus balance** (with wagering rules before bonus converts to cash)
- **Deposits:**
 - Telebirr (direct via Ethio Telecom Fabric gateway, or via aggregator)
 - Chapa / ArifPay / SantimPay aggregator checkout → covers major banks, cards, CBE Birr, M-Pesa
 - **Manual bank transfer with receipt upload + admin approval** (essential fallback — widely used in Ethiopia)
 - All using the **tenant’s own merchant credentials** — funds settle directly with the client
- **Withdrawals:** to Telebirr or bank account; minimum/maximum amounts; fee schedule; **admin review queue** with maker-checker approval; automatic prize-tax withholding where applicable
- Idempotent payment-webhook processing (a gateway retry must never double-credit)
- Daily **reconciliation screen:** tenant ledger vs. gateway settlement reports, with mismatch flags
- Full transaction history for users (deposits, plays, wins, withdrawals, bonuses)
- Refund/void flows with audit trail

4.3 Games & ticketing

- **Game engine supporting multiple templates:**
 - **Draw lottery** — pick N of M numbers (e.g., 5/35), scheduled draws, prize tiers
 - **Raffle** — fixed pool of numbered tickets, one or more winners drawn from sold tickets
 - **Instant win / scratch** — predetermined prize pool, reveal on purchase
 - Daily / weekly / special-event draws
- Game configurator (tenant admin): ticket price, sales window & cutoff, prize structure (**fixed prizes** or **pari-mutuel** % of pool), max tickets per user, rollover rules
- Ticket purchase: manual number pick, **quick pick** (random), bulk buy, multi-draw subscription (“play my numbers for the next 4 draws”)
- Ticket lifecycle: reserved → paid → active → won/lost → claimed/expired (with an unclaimed-prize policy)
- Unique ticket serials + QR code; downloadable/printable receipt
- “Check my ticket” by serial or QR scan
- Jackpot rollover and guaranteed-minimum-prize support
- Sold-ticket counter / prize-pool ticker on game pages (drives urgency, and pari-mutuel transparency)

4.4 Draw engine & provable fairness

Trust is the product — for players, for clients, and for auditors.

- **CSPRNG** (cryptographically secure random number generator) for all draws — never rand()
- **Commit–reveal scheme:** before sales close, publish the SHA-256 hash of a secret server seed; after the draw, publish the seed so anyone can verify the winning numbers were derived from it (seed + closed ticket-set hash → HMAC-based number derivation). A public “verify this draw” page recomputes it in the browser
- Draw workflow with **human approval gate:** sales cutoff → ticket set frozen & hashed → RNG executes → draw officer + second admin approve → results published everywhere simultaneously (web, bot, Telegram channel, SMS to winners)
- Automatic winner computation across all prize tiers; documented tie-handling for raffles
- **Tamper-evident audit log** — hash-chained log of every draw event
- Support for supervised/manual draws (e.g., a livestreamed physical draw for marketing) with dual-control result entry

4.5 Telegram bot (one per tenant)

- /start with **deep-link payloads** (referral codes, campaign tracking, account-link tokens)
- Registration via Telegram’s **share-phone-number** button + OTP
- Browse open games, buy tickets with inline keyboards (quick pick & manual pick)
- Wallet in chat: balance, deposit (aggregator checkout link or Telebirr flow), withdrawal request
- My tickets, results lookup, transaction history
- **Push notifications:** draw reminders, results, personal win alerts (“🎉 You won 500 ETB”)
- **Mini App button** that opens the tenant’s web app inside Telegram
- Referral link generator with reward tracking
- Amharic/English (auto-detect from Telegram locale, user-switchable)

- FAQ/support commands with handoff to a human support inbox
- Tenant-admin **broadcast tool** with opt-out handling and rate limiting (Telegram allows ~30 msgs/sec per bot — queue broadcasts)
- Auto-posting of results & jackpot milestones to the tenant's public channel

4.6 Notifications & messaging

- Channels: SMS (OTP, win alerts), Telegram push, in-app inbox, optional email
- Local SMS gateway (e.g., AfroMessage or GeezSMS) — per-message cost metered against the tenant's quota; Telegram push is free
- Templated messages, multi-language, with a per-user channel-preference matrix
- Event triggers: OTP, deposit confirmed, ticket purchased, draw reminder, results, win, withdrawal status, KYC status, promos (opt-in)

4.7 Growth & marketing

- **Referral program:** unique codes/links (web + Telegram deep links), reward on referee's first deposit/play, anti-abuse rules
- Promo codes; first-deposit / first-play bonuses (using the bonus-balance rules in 4.2)
- Loyalty points on spend, redeemable for tickets
- CMS-managed banners, landing pages, winner stories ("social proof" page — with winner consent)
- Campaign scheduler for SMS/Telegram blasts with segmentation (inactive 30 days, big players, etc.)
- UTM/campaign attribution so the client knows which channel sells tickets

4.8 Tenant admin console & operations

- **Role-based access control:** owner, finance, draw officer, support, KYC reviewer, auditor (read-only)
- **Maker-checker (two-person) approval** on all sensitive actions: draw execution, withdrawal batches, manual wallet adjustments, game config changes
- Dashboard: sales today/this draw, gross gaming revenue, active users, **outstanding prize liabilities**, deposit/withdrawal volumes
- Game & draw management (create, schedule, pause, void with reason)
- Player management: search, profile, activity, KYC review queue, suspend/ban with reason, notes
- Finance: withdrawal approval queue, manual-deposit receipt review, reconciliation, adjustment journal (dual-control)
- **Fraud tools:** velocity limits, duplicate-account detection (device fingerprint, IP, ID number), watchlists, automatic holds on suspicious wins
- Content management: FAQ, terms & conditions (versioned, with user-acceptance tracking), banners
- Immutable **audit log viewer** (who did what, when, from where) with CSV/Excel export

4.9 Reporting (per tenant)

- Revenue and prize-payout reports per game / per period, exportable (CSV/Excel)
- Winner tax-withholding report
- Large-transaction flags, structuring detection, KYC-tier enforcement

- Read-only auditor account and periodic report exports
- Consent records, retention schedule, unclaimed-prize handling

4.10 Support & content

- Help center / FAQ (web + bot)
- Support inbox: web contact form + Telegram messages land in one queue with player context attached
- Prize-claim dispute flow with status tracking
- Public pages: how it works, past results archive, draw verification, responsible gaming, T&C, privacy policy — Amharic + English

4.11 Analytics

- Funnel: visitor → registered → KYC → first deposit → first play → repeat
- Retention cohorts, average revenue per user, game-level performance (sell-through %, margin)
- Channel split: web vs. bot vs. Mini App
- Draw-night load and conversion reporting

4.12 Security & non-functional requirements

- TLS everywhere; HSTS; passwords/PINs hashed with Argon2/bcrypt
- KYC documents encrypted at rest, access-logged
- **Rate limiting + CAPTCHA on OTP endpoints** (SMS-pumping fraud is a real, expensive attack)
- Idempotency keys and DB row-locking on all money operations (no double-spend under concurrent requests)
- **Tenant isolation tests in CI** — automated proof that no query crosses tenants
- Cloudflare in front: DDoS protection, WAF, caching, hides origin IP; wildcard DNS for tenant subdomains
- Daily **off-server encrypted backups**, restorable per tenant (the wallet ledgers are the business; losing them is unrecoverable)
- Monitoring: uptime checks (UptimeRobot), error tracking (Sentry), disk/CPU alerts
- Load target: comfortable when multiple tenants hit draw-night spikes simultaneously; queue-based processing absorbs bursts
- Amharic (Ge'ez script) rendering and lightweight pages for low-bandwidth mobile users
- Staging environment; penetration test before real-money launch

5. Phased rollout

PHASE	SCOPE	CUT LINE
MVP (launch)	Multi-tenant schema from day one, operated with a single anchor tenant (manually provisioned — possibly our own lottery operation or a committed first client). One game type (raffle or simple draw lottery), phone+OTP accounts, KYC tiers, wallet with one aggregator + manual deposit fallback, commit-reveal draw engine with approval gate, per-tenant Telegram bot, tenant admin console with RBAC + maker-checker, revenue & tax reports, referral codes, Amharic+English	Everything needed to run one honest, auditable lottery end-to-end on tenant-scoped foundations
V1.5	Platform console: tenant onboarding wizard, subscription plans, billing & usage metering, feature flags. Second game type (instant win), Telebirr direct integration, Mini App polish, loyalty points, campaign broadcasts, richer analytics	Sell to the second, third, fourth client without engineering work per client
V2	Custom-domain automation, database-per-tenant premium isolation, agent/retailer portal, USSD, native apps, live-draw streaming, advanced fraud scoring, public API for tenants	Scale & differentiation

6. Recommended architecture & stack

CORE PRINCIPLE

One backend, one database, one ledger — tenant-scoped everywhere. Web, bot, and Mini App are three doors into the same system; tenants are rows, not deployments.



Recommended: Laravel 12 + MySQL or PostgreSQL + Redis

- **stanc1/tenancy** (tenancyforlaravel.com) for multi-tenancy — supports shared-DB now, DB-per-tenant later, and per-tenant domains
- Filament for both consoles: platform admin and tenant admin (RBAC, tables, approvals — very fast to build)
- Laravel Horizon for queues; scheduler for draw automation and billing metering

- Telegram bots via webhook in the same app (Nutgram or Telegraph package), one webhook route per tenant bot
- Frontend: Livewire or Inertia+Vue with per-tenant theming; the same responsive app registers as each tenant's Telegram Mini App
- Why: largest hire-able developer pool in Ethiopia, batteries included, one deployable unit on one VPS

Solid alternative: Node.js — NestJS + grammY + Prisma + PostgreSQL, Next.js frontend. Choose this only if the developers we already have are stronger in TypeScript.

7. Hosting: Yegara vs. the alternatives

The constraint that decides everything: how we pay

International providers require a Visa/Mastercard that works online in EUR/USD. From Ethiopia that means a **multi-currency prepaid card** (e.g., Dashen Mastercard, and similar from CBE/Zemen — loaded at forex bureaus under NBE rules). If we can get and reliably reload one, international hosting is far cheaper. If not, we pay in birr locally.

The candidates (like-for-like ≈ 4 vCPU / 8 GB RAM)

PROVIDER	PLAN	SPECS	PRICE/MO	PAY IN	NOTES
Yegara (ET)	VPS 80G	4 vCPU, 8 GB, 80 GB NVMe, 20 TB	4,450 ETB	Birr	Unmanaged; local support; best local value
Yegara (ET)	VPS 40G	2 vCPU, 4 GB, 40 GB NVMe, 20 TB	2,950 ETB	Birr	Viable MVP start / staging box
Zergaw (ET)	CS04008	4 vCPU, 8 GB, 80 GB	10,458 ETB	Birr	In-country DC, weekly backups — but 2 Mbps uplink standard, a serious bottleneck on draw nights
Ethio Telecom Telecloud (ET)	quote-based	varies	varies	Birr	In-country, enterprise-oriented; worth a quote if in-country hosting is required
WebSprix (ET)	varies	varies	varies	Birr	ISP with cloud/VPS offering; get a quote
Hetzner (DE/FI)	CX32	4 vCPU, 8 GB, 80 GB NVMe, 20 TB	~€7 (~1,200 ETB)	Card (EUR)	Best price/performance anywhere; superb reliability; ~130–180 ms from Ethiopia
Contabo (DE)	Cloud VPS	4 vCPU, 8 GB, 100–200 GB	~\$7	Card	Cheapest, but oversold nodes → inconsistent performance; weaker support
DigitalOcean	Basic 8 GB	4 vCPU, 8 GB, 160 GB	\$48	Card	Polished ecosystem, poor value here; no African region

Prices checked July 2026 — verify before purchase.

What actually matters for this app

- **Latency:** local hosting gives Ethiopian users ~10–40 ms vs. ~150 ms from Europe. Noticeable but fine for a lottery site behind Cloudflare caching. Telegram bot traffic routes through Telegram’s European servers anyway — EU hosting is, if anything, *better* for bot responsiveness.
- **Payment webhooks** from Telebirr/Chapa reach either location without issue.
- **Data residency:** confirm whether player data must be hosted in Ethiopia before committing — only the Ethiopian providers qualify if so.
- **SaaS scaling:** tenants share the same box; each active tenant adds modest load. Plan to step up tiers as tenants sign — Hetzner’s 5-minute resize/snapshots make this much easier than local providers’ manual upgrades.
- **Reliability:** Hetzner’s track record beats any local option; among locals, Yegara’s VPS line (NVMe, 20 TB transfer) is markedly better value than Zergaw’s pricing and 2 Mbps uplink.

Recommendation

1. **If offshore hosting is allowed and we can get a multi-currency card** → **Hetzner CX32** (~1,200 ETB/mo). Same specs as Yegara 80G at 1/4 the price, plus snapshots and a proper cloud firewall. Keep the domain purchase at Yegara (pay in birr) or Namecheap.
2. **If we must pay in birr or host in Ethiopia** → **Yegara VPS 80G** (4,450 ETB/mo). Best local value by a wide margin.
3. **Either way:** Cloudflare free tier in front with wildcard DNS for tenant subdomains; a small second box (Yegara 40G or Hetzner CX22 ~€4) as staging + off-site backup target.
4. **Avoid shared hosting** for the app itself: no queue workers, no Redis, no root hardening, no wildcard subdomain control, shared IP — wrong tool for a real-money multi-tenant platform. (A cheap Yegara shared plan is fine for a marketing/landing site + email if wanted.)

8. Supporting services

SERVICE	CHOICE	COST
DNS / DDoS / WAF / CDN	Cloudflare free tier (wildcard *.ourplatform.com supported)	0
SSL	Let's Encrypt (auto-renew; wildcard via DNS challenge + per-custom-domain issuance)	0
SMS gateway	AfroMessage / GeezSMS	per message (~0.3–0.6 ETB), metered per tenant — use Telegram push wherever possible
Payment aggregator	Each tenant connects their own Chapa / SantimPay / ArifPay / Telebirr merchant account	tenant pays ~2–3% per transaction to their gateway (confirm gateways will onboard lotteries)
Error tracking	Sentry free tier	0
Uptime monitoring	UptimeRobot free	0
Backups	Nightly encrypted DB dump → second VPS/off-site (or Backblaze B2 if paying by card, ~\$6/TB); per-tenant restore tested	~0–1,000 ETB
OS & stack	Ubuntu LTS, Nginx, firewall (80/443/SSH only), fail2ban, SSH keys only	0

9. Cost summary

One-time (technical): development (in-house partner effort or contracted) and a pre-launch penetration test. Company setup, client contracts, and licensing costs are tracked separately — each tenant carries their own operating license and prize funding.

Monthly infrastructure (platform side):

ITEM	BIRR-ONLY ROUTE	INTERNATIONAL-CARD ROUTE
Production server	Yegara 80G — 4,450 ETB	Hetzner CX32 — ~1,200 ETB
Staging/backup box	Yegara 40G — 2,950 ETB (optional at start)	Hetzner CX22 — ~650 ETB
Domain (.com or .com.et)	~150–250 ETB/mo amortized	same
SMS budget	usage-based, metered per tenant	same
Total infrastructure	~5,000–7,500 ETB/mo	~2,500–4,000 ETB/mo

Tenant-variable costs (SMS beyond quota, payment-gateway fees) pass through to each client's own accounts. Platform infrastructure steps up roughly per handful of active tenants — and each new tenant should more than cover it in subscription revenue.

10. Decisions to make together

1. **Anchor tenant** — launch with our own lottery operation as the first tenant, or a committed first client?
2. **SaaS pricing model** — setup fee + monthly subscription + revenue share (% of ticket sales) vs. per-ticket fee; trial policy; what the plan tiers gate (game types, player caps, SMS quota, custom domain)
3. **Target clients** — existing betting shops adding lottery, new lottery entrants, corporate/NGO promotional raffles (a lighter niche worth exploring)
4. **First game type** — raffle (simplest to explain and audit) vs. pick-N draw lottery (rollover jackpots)? Fixed prizes vs. percentage-of-pool?
5. **Hosting route** — can we obtain a multi-currency prepaid card (→ Hetzner), or birr-only (→ Yegara 80G)?
6. **Payment partners to support first** — Chapa vs. SantimPay vs. ArifPay? (Ask each about onboarding lotteries + settlement timelines)
7. **Stack & team** — Laravel (recommended for local hiring) vs. Node.js; who builds, and who runs platform operations (tenant onboarding, billing, support)
8. **Platform brand & domain** — each client brings their own brand; secure the platform name and matching Telegram handles early

Sources

- Yegara plans & pricing: [whotop listing](#), yegara.com/vps, yegara.com/price
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- Multi-tenancy: [Tenancy for Laravel](#) · Lottery software feature references: [GammaStack](#), [Innosoft](#)

Prices in this document are from public sources as of July 2026 and must be re-verified with banks and providers before commitments are made.